

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	9 February 2026
Team ID	LTVIP2026TMIDS36203
Project Name	Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization
Maximum Marks	4 Marks

Technical Architecture

This solution follows a layered architecture combining data analytics and web integration to analyze product placement impact on sales. The architecture ensures modular design, interactive visualization, and smooth deployment through web technologies. The system flow includes data preparation, visualization development in Tableau, and dashboard embedding using a Flask web application. Sales datasets are cleaned and structured before being imported into Tableau for analysis. Interactive dashboards are created to visualize sales trends, regional performance, and placement effectiveness. The Tableau dashboard is then embedded into a Flask-based web interface, allowing users to access insights through a browser. This architecture ensures scalability, performance efficiency, and ease of access while maintaining a clear separation between data processing, visualization, and presentation layers.





Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web interface through which users access the sales dashboard	HTML, CSS, Flask
2	Visualization Layer	Interactive dashboard displaying sales trends and placement insights	Tableau Desktop / Tableau Public
3	Application Logic	Handles routing and rendering of web pages	Python (Flask Framework)
4	Data Processing	Cleaning and structuring sales dataset before visualization	Excel / CSV Processing
5	Deployment Layer	Hosting and embedding dashboard for user access	Tableau Public + Flask Local Server

 **Table-2: Application Characteristics**

S.No	Characteristics	Description	Technology	
1	Open-Source Framework	Flask is a lightweight open-source Python web framework	Flask	
2	Security Implementation	Secure dashboard access when deployed; public sharing via Tableau Public	Flask Routing, Tableau Public	
3	Scalable Architecture	Can support additional datasets and advanced analytics	Modular Dashboard Design	
4	Availability	Accessible via web browser anytime	Web-based Deployment	
5	Performance	Optimized dashboard loading with interactive filters and efficient data handling	Tableau Engine + Flask Server	